Research Square

Preprints are preliminary reports that have not undergone peer review. They should not be considered conclusive, used to inform clinical practice, or referenced by the media as validated information.

Adoption of drone technology in the Indian construction industry

Sivaraman P

sivaraman080bem23@spa.ac.in

School of Planning and Architecture Delhi

Luke Judson

School of Planning and Architecture Delhi

Research Article

Keywords: Drones, Unmanned Aerial Vehicle (UAV), Indian construction, Application, Success Factors

Posted Date: May 15th, 2025

DOI: https://doi.org/10.21203/rs.3.rs-6470345/v1

License: This work is licensed under a Creative Commons Attribution 4.0 International License. Read Full License

Additional Declarations: No competing interests reported.

Abstract

Drones have emerged as a widely adopted digital technology across various industries, offering transformative advancements in construction. Application of drones are expanded throughout all phases of constructions, including pre-construction site analysis, active construction management, and post construction documentation. In the Indian context, the drone market is expected to grow at a compound annual growth rate (CAGR) of 80% from 20 to 25; however, its adoption remains predominantly concentrated in military and agricultural applications. While previous research has mainly focused on the operational-level implementation of drones in construction, limited studies have examined their strategic adoption. This paper addresses this gap by investigating the factors influencing drone adoption at a strategic level. A survey-based research approach was proposed to collecting data from key stakeholders in the Indian construction industry. Statistical analyses, including binary regression modelling, were conducted to assess the impact of economic, social-legal, operational, and construction-related factors on drone adoption. The findings highlight those economic advantages such as project costs and optimized resource utilization, as primary drivers of drone adoption. Operational factors, including access to high-quality equipment and training programs, are also critical. Furthermore, Social acceptance and compliance with government regulations are essential, along with meeting the construction industry's demand for precise data collection and efficient inventory management. To overcome existing barriers, this study recommends targeted investments in skill development, policy interventions to streamline regulatory frameworks, and incentives for domestic drone manufacturing. Integrating drones with advanced construction technologies is essential to enhancing efficiency, data-driven decision-making, and overall industry transformation. These strategic measures are vital for unlocking the full potential of drones in India's construction sector.

1. Introduction

The construction industry is growing day by day, and construction sites and tasks are becoming increasingly complex and diverse. Because a large portion of the workforce is transitory, annual productivity growth is only one-third of typical economic growth (Li & Liu, 2018). This leads to the introduction of automation and digital technologies to enhance productivity (Alziadi & Qasem, 2022). In building projects, the modern technologies are used for speed, precision, and safety. Modern, cutting-edge technology are being introduced into the construction sector to expedite project completion. Monitoring techniques are also being updated to keep up with the speed of execution.

An Unmanned Aerial Vehicle (UAV) or Drone is a small aircraft without a human pilot aboard. Drones consist of a control system, rotor, motor, cameras, landing gear, altimeter and communication system. Initially, Drones were used only for military purposes (Gulshan Taj, et al., 2021). But in recent times, due to its remarkable improvement in software and hardware components like increased battery life for longer flights, heavier payloads to hold more sensors, and improved signal ranges, this new technology is finding a predominant place in the construction industry, making the work faster at low cost (Vanathi & Radhika, 2022).

Drones are a crucial project management tool in building projects. By taking high-resolution pictures and videos, drones help with visual inspection procedures like surveying, area mapping, following the progress of construction projects, on-site equipment and material tracking, building inspections, spotting construction flaws, and 3D modelling (Kumarapu, et al., 2020). Drone technology can be used throughout the lifecycle of a project (pre-construction, during construction & post construction) to record and document valuable information and progress. The emerging new drone inspection technologies like LIDAR (Light Detection and Ranging), SLAM (Simultaneous Localization and Mapping), Laser scanning, photogrammetry and Thermography, along with processing software, provide precise real-time scenarios (Sridharan, et al., 2023). Drone maps can also be geotagged, allowing for the instantaneous estimation of stockpile volumes for in-the-moment decision-making and basic area measurements. Drone data integration with BIM (Building Information Modelling) assists in critical analysis of a project by comparing as-built vs as-designed. Recent studies show the incorporation of RFID (Radio Frequency Identification) technology along with Drones, which aids in identifying equipment and materials in construction sites (Nachiappan, et al., 2024).

In 2018, the global construction sector saw a 239% increase in the adoption of Drone technology, according to the (PWC, 2023). The report states that drones help a construction project by providing increased efficiency and accuracy of a project and eliminating miscommunication in a project at a given point in time. The global drone market is poised to become a US$ 54 Billion market by 2025. Technology is having a major influence on the construction business in India, which is going through a massive transition. In the last few years, the usage of drones in construction management has grown (Li & Liu, 2018). Drones are becoming one of the most interesting developments in building. Drone use in construction is expected to increase over the next ten years, and they will be essential to futuristic buildings. In the Indian economy, the construction sector is among the most distinctive, adaptable, and active. The Indian drone market has seen an increase of 18% CAGR (Compounded Annual Growth Rate) from 2022 to 2025 because of major usage in other disciplines like military and agriculture. Still, the majority of the Indian construction industry is lagging in the usage of Drones. Drone in the construction industry increases cost savings by 5-20%, 52% of the time of data insights, 61% of more accurate measurements, 55% of increased safety, and 65% of improved communication and collaboration (PWC, 2023). The Indian government have taken initiatives to expand access to the PLI scheme for drones and drone components for providing subsidies and promoting investments from micro, small, and medium-sized enterprises. The drone manufacturing potential in India could be worth US$ 4.2 Billion by 2025 (EY, 2022). There are significant safety and legal considerations, and users should understand the distinctions between commercial and personal use of drones. All drones must be FAA registered. Despite increased usage and regulation, the use of drones in construction continues to evolve significantly (Fan & Saadeghvaziri, 2019). Central Government projects have already started using Drone technology (e.g. Statue of Unity, Light House Projects, etc.), NHAI (National Highway Authority of India), in July 2021, made Drone monitoring mandatory for all future highway projects for monitoring different stages of development, construction, operation and maintenance. It indicates that the application of drones has already been explored in Indian construction sites (Vanathi & Radhika, 2022).

Although the benefits of this technology in construction appear unlimited, its integration into organizational activity is strongly dependent on the balance of its advantages and limitations (Charlesraj & Rakshith, 2020). Organizations in the construction sector are less prepared for drones than those in the agricultural or logistics sectors. There is a disconnect between drones' potential benefits and their actual application in the building sector. (Albeaino & Gheisari, 2021).

Researchers and practitioners have tried to use drones for commercial reasons in a variety of industries, according to the literature. Because drones may save money, they have a big impact on the building industry (Judson & Paul, 2019), reduced time, accurate measurements, and increased safety (Falorca, et al., 2021). However, a number of authors have noted that the lack of government regulations pertaining to drone uses and issues with people's privacy and safety may make it more difficult for drones to be used in civic applications (Raj & Sah, 2019).

It's interesting to note that all of the above research was done at the operational level, no strategy study has been done to examine the success variables influencing drone adoption in the construction industry. (Raj & Sah, 2019). Several authors have analyzed success factors in different contexts, such as sustainability initiatives in supply chains (Luthra et al., 2021), traceability for food logistics systems (Shankar et al., 2023), new product development (Cooper & Kleinschmidt, 1995), and ERP planning (Mashari et al., 2003). However, no researcher has analyzed success factors for the adoption of drones in the construction sector. So, the aim is to bridge this gap by investigating the influential success factors for the better adoption of drone technology in the Indian Construction Industry (Patil, et al., 2022).

2. Literature Review

In the recent years, there has been a rapid development of unmanned aircraft equipped with a remote control called Unmanned Aerial Vehicles (UAV), commonly known as Drones in the world. This equipment is well-designed and easy to control. So, it became popular around the world. UAVs or drones can be classified according to their usage and application. In the early 19, drones were used for military purposes. Nowadays, Drones are used for several purposes, especially in media-related works, Agricultural purposes, Mining purposes, surveying works, and Construction industry (Sankalpa, 2020).

Drones can be classified as fixed-wing, single-rotor, and multirotor. When compared to conventional UAV frameworks, multirotor drones, such as quadcopters, have unique advantages, for example, strength, high mobility, and low buy and upkeep costs. Multirotor drones have multiple rotors and utilize fixed-pitch sharp edges (Kaamin, et al., 2023). By varying the total speed of each rotor, the push and torque that each one delivers can be altered, allowing for control over the movement of the vehicle. Multirotor drones can be moved in little spaces while drifting and can be constrained by different gadgets, for example, tablets, PCs and personal computers. They can likewise be effectively furnished with light discovery and extending (LIDAR) instruments, cameras, and specialized gadgets. In this way, numerous

fields indicate expanding enthusiasm for using multirotor drones for different non-military purposes (Li & Liu, 2018).

Process of drone technology

Drone technology enhances construction productivity by enabling efficient data collection and analysis. Drones, equipped with advanced surveying and detecting systems, support various project phases, including planning, design, construction, and maintenance (Rao, et al., 2022). UAVs, classified into fixed-wing, rotary-wing, and hybrid types, offer unique advantages, such as extended flight endurance, easy maneuverability, and high-resolution imaging (Elghaish, et al., 2020). Compared to conventional aerial platforms, drones provide cost-effective solutions by operating at low altitudes, accessing hard-to-reach areas, and eliminating the need for licensed pilots (Falorca, et al., 2021). Their detection and surveying capabilities include high-resolution cameras, thermal imaging, LiDAR, and RADAR, improving site monitoring and structural assessments. Post-data processing integrates machine learning and deep learning techniques for real-time construction monitoring and defect detection, with pre-trained models achieving high accuracy in identifying cracks, corrosion, and structural deformations (Rajagopalan & Krishna, 2018).

Indian drone regulation

India classifies Unmanned Aircraft Systems (UAS) into aero planes, rotorcraft, and hybrid systems, further categorised by weight: Nano (<250g), Micro (250g–2kg), Small (2–25kg), Medium (25–150kg), and Large (>150kg) (DGCA, 2018). All drones except Nano require registration with a Unique Identification Number (UIN), and commercial operations need a permit, except for Microdrones flying below 200 feet. Flights must remain within visual line of sight and below 400 feet, avoiding restricted zones such as airports, borders, and military sites. India’s "No Permission, No Takeoff" (NPNT) policy mandates pilots to seek flight approval via the Digital Sky Platform before every takeoff. Drone registration involves submitting specifications on the platform, obtaining a UIN, and acquiring a Remote Pilot Certificate through DGCA-approved training. Foreign drone imports require DGCA clearance, adhering to NPNT rules and India's technical standards to ensure safe airspace management (The Drone rules, 2021).

Sensing technologies

Drones also include various sensors tailored to their applications in construction and data collection. Distance sensors such as LiDAR, ultrasonic, radar, infrared (IR), Time-of-Flight (ToF), and stereo vision provide capabilities for mapping, obstacle detection, and precise surveying. Magnetic field sensors like magnetometers and Hall Effect sensors assist in navigation, motor control, and infrastructure inspection. Orientation sensors, including gyroscopes, accelerometers, inertial measurement units (IMUs), and barometers, ensure stability, altitude control, and smooth flight, which is essential for aerial surveys and detailed inspections. The application of these sensors with their cost and accuracy are mentioned in the Table 1 below. These sensors collectively enable drones to perform efficiently in

complex construction environments, enhancing safety, accuracy, and operational flexibility (Rao, et al., 2022).

Table 1 Types of Sensors and their accuracy (Source: Author)

Sensor Type	Applications in Construction	Brand & Model	Cost	Accuracy	Reference
LiDAR	High-resolution 3D mapping, terrain modelling, and structural inspections.	Velodyne Puck VLP-16	₹ 8,75,000	±5 mm (High)	(Ouster)
Ultrasonic	Short-range obstacle detection, indoor navigation, altitude stabilization.	Max Botix MB7389	₹ 8,500	±5 cm (Moderate)	(Max botix)
Radar	Long-range detection in poor visibility (fog, dust), obstacle avoidance.	Texas Instruments AWR1843BOOST	₹ 22,500	±1 m (Moderate to High)	(Texas Instruments)
Infrared (IR)	Indoor inspections temperature difference detection (e.g., heat leaks).	FLIR Lepton 3.5	₹ 7,000	±1 m (Moderate)	(Flir)
Time-of-Flight (ToF)	Altitude measurement, terrain following, obstacle detection.	STMicroelectronics VL53L1X	₹ 1,200	±1 cm (High)	(st)
Stereo Vision	3D mapping, obstacle avoidance, visual progress tracking.	Intel RealSense D435	₹ 15,000	±5 cm (High)	(Intel real sense)
Magnetometers	Navigation, maintaining stable course over large areas.	Honeywell HMC5883L	₹ 2,300	±2° (Moderate)	(Honeywell)
Hall Effect	Motor speed regulation, detecting metallic infrastructure.	Allegro Microsystems A3144	₹ 300	Moderate	(Amazon)

Gyroscopes	Stabilization during aerial surveys and inspections.	InvenSense MPU6050	₹ 300	±0.1° (Moderate)	(Invensense)
Accelerometers	Precise maneuvers, hovering, and smooth navigation over uneven terrain.	Analog Devices ADXL345	₹ 500	±0.1 m/s² (High)	(Analog)
Inertial Measurement Unit (IMU)	Critical for aerial inspections, mapping, and progress monitoring.	Bosch BNO055	₹ 4,000	High	(Bosch)

Application of drone technology

Table 2 Details of various construction stages, along with their applications of drones (Source: (Mahajan, 2021)

Construction Stages			
Stage		**Application**	**Reference**
I -Pre-construction stage	Site Planning	Suitable site Selection Topographic survey Zoning	*(J.John, et al, 2023), (A. Davis, 2021)*
	Property Purchase & decision making	Site location Surrounding Analysis Site barriers	*(Z. Dukowitz, 2020), (J. Patterson, 2018), (M. Sobola, 2021)*
II-During construction stage	Earthwork estimation & volumetric analysis	Photogrammetry output Digital terrain model	*(PaeNatwilai, 2021), (Jezekiel R. Bognot et al, 2018), (J.Sanchez, 2019)*
	Construction Inspection	Concreting, material transportation, fabrication, installation work ready mix concrete plant Orthomosaic	*(Ruiz et al., 2021), (P. Maranpour., 2019), (D.Giordan et al., 2020), (R.Ashour et al., 2022), (Treck.R et al., 2018), (P.Nousi et al., 2019)*
	Progress Tracking	Real-time work progress Decision making Handle multiple sites	*(P.Nousi et al.,2019), (I.Barnes., 2020), (Seogdeok et al., 2017)*
	Labour safety & surveillance	Safe construction work Labour productivity Monitoring, surveillance & tracing labour's activities	*(U. Patel,2021), (J. Howard, 2023), (Tuttas S et al., 2021), (W.Yi et al., 2020),* (Vishwakarma, et al., 2022)
	Actual site data measurement	Precise aerial photogrammetry 2D & 3D for correct measurement Site assets using Orthomosiac Analysis of masonry structures	*(Wintra et al., 2018), (N.atwal., 2021),* (Yadav, et al., 2022)
III-Post -Construction Stage	Maintenance & Defect Analysis	Feasible for data collection from large and multi-storied structures Minimise error during operation & maintenance Fire risk analysis for construction sites	*(B.Vergouw et al, 2022), (A.Lin et al., 2018),* (Simon, et al., 2022)
	Marketing & Promotion	Provides promoters and clients with a visual demonstration of actual site work progress Possible through videos & photos	*(A.Lin, 2023), (O.Maghazel et al., 2020)*
IV-Closeout	Project Closure	Evaluation of work from start to finish	*(S.Tasevki et al., 2018)*

Drones are increasingly being integrated across various stages of construction, offering numerous applications that enhance accuracy, safety, and efficiency. In the pre-construction stage, drones assist in site planning by enabling better site selection through topographic surveys and zoning analysis (Zhang, et al., 2023). They also support informed property purchase decisions by providing detailed insights into site location, surroundings, and potential barriers (Dukowitz, 2020; Patterson, 2018; Sobola, 2021). During the construction phase, drones facilitate accurate earthwork estimation and volumetric assessments through photogrammetry and digital terrain modeling (Li & Liu, 2018). A study conducted by (Ciampa, et al., 2016) for construction inspection, drones aid in monitoring processes such as concreting, material transport, and on-site fabrication, including visual assessments using Orthomosaic imagery (Ruiz et al., 2021; Marzappour, 2019; Giordan et al., 2020; Ashour et al., 2022; TreckR et al., 2018; Nousi et al., 2019). They are also instrumental in progress tracking, offering real-time updates, aiding decision-making, and improving coordination across multiple tasks (Saini, et al., 2021).

Furthermore, drones play a vital role in ensuring labor safety and enhancing surveillance by tracking workforce productivity and monitoring site activities (Patel, 2021; Howard, 2023; Tutas et al., 2021; Yi et al., 2020; Vishwakarma et al., 2022). They contribute to precise site data measurement through high-resolution aerial imagery, 2D and 3D mapping, and analysis of structural elements (Liang, et al., 2023). In the post-construction phase, drones help in defect detection and maintenance analysis, particularly for complex structures, by minimizing errors and assessing fire risks (Vergouw et al., 2022; Lin et al., 2018; Simon et al., 2022). For marketing and promotion, drones provide visual demonstrations of project progress through imagery and videos (Rao, et al., 2022). Finally, during project closeout, they assist in evaluating completed work from initiation to completion (Tasevski et al., 2018). The Table 2 shows a description of the four main stages and their potential in detail (Mahajan, 2021).

Factors Influencing the Adoption

Economic Factors: The initial procurement cost of drones varies based on features and capabilities, with high-end models posing a financial challenge, especially for SMEs (Zhang, et al., 2023). Operational and maintenance expenses, including battery replacements, software updates, and repairs, further impact feasibility (Stocker, et al., 2020). However, drones offer cost savings over conventional methods by improving efficiency and reducing labour costs, making them attractive for data-intensive tasks. Budget availability plays a crucial role, with organizations needing financial flexibility or external funding for adoption. Additionally, data processing costs, including downloading, analysis, and interpretation, require specialized software and expertise, adding to the overall expense (Jacobsen & Teizer, 2022).

Operational Factors: While drones are increasingly accessible due to technological advancements, specialized models with high-resolution cameras, thermal sensors, or LiDAR remain limited in certain regions due to supply constraints and import regulations (Stocker, et al., 2020). Type, quality, range, and payload capacity significantly impact performance, (Moza & Paul, 2024) with long-range drones suited for large sites and smaller drone’s ideal for confined spaces. Airspace restrictions and permit requirements create regulatory hurdles, with compliance varying by country. Training and capacity building are essential for safe operation and effective data utilization but add to costs (Martinez & Gheisari, 2020). The availability of skilled human resources remains a challenge, especially in regions with limited training programs (Shakhatreh & Sawalmeh, 2019). Additionally, pilot competency is crucial for regulatory compliance and operational efficiency, as untrained operators increase the risks of accidents and inefficiencies (Zhang, et al., 2023).

Social-Legal and Environmental Factors: Government regulations impose strict requirements on drone operations, including permits, altitude limits, and maintaining line-of-sight, with non-compliance leading to penalties (Stocker, et al., 2020). Company acceptance varies, with larger firms adopting drones more readily due to resources, while smaller companies may hesitate due to cost and unfamiliarity. Piloting licenses are mandatory in many countries, adding training and exam costs that may deter adoption (Shakhatreh & Sawalmeh, 2019). While drones create employment opportunities for skilled professionals like operators and data analysts, companies must invest in training. However, automation raises

concerns about job displacement, especially for tasks like surveying (Albeaino & Gheisari, 2021). Safety risks from drone malfunctions necessitate strict protocols to protect workers. Additionally, airspace encroachment and privacy concerns in urban areas require careful regulatory compliance to address community objections (Zhang, et al., 2023).

Construction Demand-Related: Drones significantly enhance data collection by providing precise geospatial information for site surveys, 3D modelling, and progress monitoring (Zhang, et al., 2023). They streamline stocktaking and resource control, enabling accurate tracking of materials and equipment to align with project timelines (Stocker, et al., 2020). Crime and safety surveillance is improved through real-time monitoring, deterring theft and identifying hazards. Additionally, drones optimize the deployment of materials, workforce, and equipment, offering project managers a comprehensive site overview to improve planning (Albeaino & Gheisari, 2021). Their ability to deliver high-accuracy data reduces human error in surveying and workflow management, improving efficiency (Shakhatreh & Sawalmeh, 2019). Moreover, during crises like the COVID-19 pandemic, drones enable remote monitoring, ensuring continuous project oversight with minimal on-site personnel. The Table 3 explains about the combined information's of the factors and their corresponding subfactors influences the usage of the drones.

Table 3 Influencing factors for the usage of drones (Source: Author)

Factors	Subfactors	References
Economic	Cost of Procurement of Drone	*(Zhang et al., 2019)*
	Cost of operation & maintenance	*(Shakhatreh et al., 2019), (Stocker et al., 2020)*
	Cost savings relative to conventional methods	*(Siebert & Teizer et al., 2022)*
	Availability of budget	*(Dron & Dron et al., 2018)*
	Cost of data and analysis	*(Siebert & Teizer et al., 2022), (Zhang et al., 2019)*
Operational	Availability of equipment (Drones)	*(Stocker et al., 2020)*
	Type, quality, and capacity	*(Siebert & Teizer et al., 2022), (Zhang et al., 2019)*
	Air space restriction	*(Stocker et al., 2020), (McNabb et al., 2017)*
	Training and capacity building	*(Zhou & Gheisari et al., 2021)*
	Availability of Human resources	*(Zhou & Gheisari et al., 2021), (Shakhatreh et al., 2019)*
	Competency of pilots	*(Stocker et al., 2020)*
Social-legal and environmental	Government regulations	*(Stocker et al., 2020), (McNabb et al., 2017)*
	Acceptability of Drone technology by companies	*(Zhou & Gheisari et al., 2021)*
	Piloting license	*(McNabb et al., 2017), (Shakhatreh et al., 2019)*
	Employment opportunities for skilled professionals	*(Irizarry & Costa et al., 2022)*
	Threat to the Common Workforce	*(Zhou & Gheisari et al., 2021)*
	Risk and safety issues for human resources	*(Stocker et al., 2020), (Shakhatreh et al., 2019)*
	Encroachment on air space and visual incursions	*(Zhang et al., 2019)*
Construction demand related	Data/information collection	*(Zhang et al., 2019)*
	Stocktaking, control and monitoring	*(Irizarry et al., 2016), (Stocker et al., 2020)*
	Surveillance of crime and safety	*(Zhang et al., 2019)*

	Deployment of material and equipment	*(Zhou & Gheisari et al., 2021)*
	Collection of data from inaccessible	*(Zhou & Gheisari et al., 2021)*
	Data Accuracy & High-quality information	*(Shakhatreh et al., 2019)*

3. Research Methodology

For data collection, participants were chosen from various sectors of the Indian construction industry. Given that drone application in construction is still in its early stages, obtaining comprehensive and well-structured data proved to be a significant challenge. To address this, a perception-based survey was designed and distributed to gather insights from industry professionals. The survey focused on understanding the barriers, challenges, and key factors influencing drone adoption.

Participants included experienced stakeholders actively involved in construction projects as well as specialists familiar with drone technology. Selection was based on their professional background and expertise in areas such as construction management, technology integration, research, and related disciplines. In total, 50 responses were obtained from qualified professionals. Due to the emerging nature of drone use and the niche expertise required, identifying and reaching out to willing respondents was particularly difficult. It's important to note that an estimated 25,000 professionals in India are engaged in major construction firms, working across roles in engineering, design, technology, project management, and safety.

Efforts were made to include a balanced mix of professionals—ranging from contractors and consultants to academics—to ensure diverse and comprehensive viewpoints. The original sample targeted 90 individuals at a 95% confidence level, acknowledging the limited accessibility to suitable respondents. Despite the seemingly modest sample size, it was deemed appropriate given the exploratory nature of the study and the challenges faced in participant recruitment, in line with standards suggested by Data Analysis Australia (2018) and Dithebe et al. (2019).

The survey participants comprised project managers, consultants, academic researchers, contractors, and drone technology practitioners. A structured questionnaire was used to gather their views. The questionnaire covered a broad range of themes including barriers to drone usage, influencing factors, and potential strategies to support its adoption in construction. Since specific academic studies focused on the Indian context are limited, the questionnaire was developed based on available literature and grouped the influencing factors into five categories: economic, operational, socio-legal, environmental, and construction demand (Wingtra et al., 2021; Aiyetan & Das, 2023). Respondents rated each item using a five-point Likert scale where 1 represented "very low" and 5 represented "very high" significance.

The collected responses were first assessed for consistency and reliability. Standard deviation (SD) values within the range of 0.6 to 1.2 were used to evaluate consistency. Additionally, reliability was tested using Cronbach's alpha, with acceptable values ranging between 0.7 and 0.9. To measure the perceived impact of various barriers and challenges, Perception Index (PI) scores were calculated using the mean values derived from the Likert responses. Each item was also tested for statistical significance using a Z-test at a 95% confidence level. A p-value of ≤ 0.05 indicated statistical significance. Items with a PI ≥ 4 and $p \leq 0.05$ were considered highly impactful, while those scoring between 3 and 4 with significant p-values were treated as moderately influential. Factors with a PI below 3 or a p-value above 0.05 were considered to have minimal or no influence.

To further explore the factors and assess strategies that could enhance drone integration in construction, logistic regression modeling was applied. While a variety of analytical techniques could have been used, logistic regression was chosen due to the ordinal nature of the survey data and the limited availability of structured datasets. This method is particularly effective in analyzing the relationship between a dependent variable and one or more independent variables (Tutz, 2022). It allows for interpreting the strength of influence of multiple predictors on an outcome, assuming a causal or dependent link (Lu et al., 2019; Williams & Quiroz, 2020). Logistic regression also helps estimate the probability of adoption based on the influence of different factors, making it well-suited for the goals of this study. All statistical modeling and analysis were carried out using IBM SPSS software.

4. Results & Discussions

The sample size processing summary is mentioned in the table 4. The analysis included 52 valid cases, representing 100% of the sample, with no exclusions. This indicates complete data availability from all respondents, allowing for comprehensive analysis without data imputation or adjustments.

Table 4 Case processing summary of the survey

Summary			
		N	**%**
Cases	Valid	52	100.0
	Excluded	0	.0
	Total	52	100.0
a. Listwise deletion based on all variables in the procedure.			

Reliability (Cronbach's Alpha): The Cronbach's Alpha coefficient for the survey instrument is 0.878, based on 35 items is mentioned in the reliability statistics in table 5. This value falls within the accepted range of 0.7 to 0.9, commonly regarded as an indicator of high internal consistency. Such a coefficient

suggests that the items within the survey are reliable and cohesive in measuring the intended constructs.

Table 5 Reliability analysis of the survey

Reliability Statistics	
Cronbach's Alpha	N of Items
.878	35

Economic Factors

Table 6 Economic factors - Logistic regression Analysis

	B	S.E.	Wald	Sig.	Exp(B)	95% C.I.	
						Lower	Upper
Cost of Procurement of Drone	-0.130	0.543	0.057	0.811	0.878	0.303	2.547
Cost of operation and maintenance	-0.943	0.647	2.126	0.145	0.390	0.110	1.384
Cost savings relative to conventional methods	0.457	0.355	1.655	0.198	1.579	0.787	3.166
Availability of a dedicated budget	-0.072	0.416	0.030	0.862	0.930	0.412	2.100
Cost of data and analysis	0.167	0.356	0.220	0.639	1.182	0.588	2.375
Constant	3.100	2.011	2.377	0.123	22.206		

The logistic regression analysis of economic factors (Table 6) reveals that perceived cost savings when compared to conventional methods significantly encourage the adoption of drones, as indicated by a regression coefficient (B) of 0.457 and an odds ratio (Exp(B)) of 1.579. This reflects a 57.9% higher probability of adoption, suggesting that when organizations recognize notable financial benefits, they are more likely to integrate drones into their operations. Similarly, expenses related to data downloading, analysis, and interpretation show a B coefficient of 0.167 and an odds ratio of 1.182, implying an 18.2% increase in adoption likelihood, indicating that manageable data costs can have a modest positive effect on adoption.

Conversely, the initial investment required for drone procurement is associated with a B coefficient of -0.130 and an odds ratio of 0.878, reflecting a 12.2% decrease in the probability of adoption, which suggests that high upfront costs may act as a minor deterrent. The operational and maintenance expenses show a much stronger negative impact, with a B value of -0.943 and an odds ratio of 0.390, indicating a 61.0% decline in adoption likelihood. This underscores that ongoing costs such as maintenance, updates, and servicing pose a significant challenge. Furthermore, the lack of a dedicated

financial allocation for drone technology, with a B value of -0.072 and an odds ratio of 0.930, is associated with a 7.0% reduction in adoption chances, implying a relatively lesser influence compared to other economic constraints.

Operational Factors

Table 7 Operational factors - Logistic regression Analysis

	B	S.E.	Wald	Sig.	Exp(B)	95% C.I.	
						Lower	Upper
Availability of drone equipment	0.023	0.545	0.002	0.967	1.023	0.352	2.976
Type, quality, and capacity	-1.423	0.528	7.266	0.007	0.241	0.086	0.678
Air space restriction	-0.192	0.413	0.216	0.642	0.825	0.367	1.855
Training and capacity building	0.739	0.472	2.451	0.117	2.094	0.830	5.281
Availability of Human resources	0.307	0.447	0.470	0.493	1.359	0.565	3.265
Competency of drone pilots	-0.273	0.472	0.333	0.564	0.761	0.302	1.922
Constant	4.368	2.405	3.299	0.069	78.892		

The logistic regression analysis of operational factors (Table 7) indicates that training initiatives and capacity-building efforts play a crucial role in promoting drone adoption. A regression coefficient (B) of 0.739 and an odds ratio (Exp(B)) of 2.094 suggest that enhanced training availability and effectiveness more than double the probability of adoption (by 109.4%). This underscores the significance of structured programs to equip personnel with the necessary skills and boost organizational confidence in drone technology. Likewise, the presence of adequate human resources, reflected by a B value of 0.307 and an odds ratio of 1.359, leads to a 35.9% rise in adoption probability, pointing to the positive influence of workforce availability. Meanwhile, access to drone equipment shows only a marginal effect, with a B value of 0.023 and Exp(B) of 1.023, corresponding to a 2.3% increase in likelihood, suggesting that mere availability of hardware is not a decisive factor in driving adoption.

In contrast, technical specifications of equipment - such as type, quality, and capacity act as a major limiting factor. This is evident from a B coefficient of -1.423 and an odds ratio of 0.241, indicating a 75.9% decrease in the likelihood of adoption, which highlights the critical need for reliable and high-performance drones. Similarly, airspace restrictions and the requirement for flight permissions present operational and regulatory challenges, as shown by a B value of -0.192 and an odds ratio of 0.825, resulting in a 17.5% decline in adoption likelihood. Finally, insufficient pilot competency emerges as a moderate barrier, with a B value of -0.273 and an odds ratio of 0.761, reducing adoption chances by 23.9%, implying that enhancing pilot training could alleviate this constraint.

Social, Legal & Environmental Factors

Table 8 Social & Environmental factors - Logistic regression Analysis

	B	S.E.	Wald	Sig.	Exp(B)	95% C.I.	
						Lower	Upper
Government regulations and policies	0.041	0.473	0.007	0.932	1.041	0.412	2.632
Acceptability of Drone technology by companies	-0.862	0.420	4.201	0.040	0.422	0.185	0.963
Requirements for obtaining a Pilot license	0.029	0.428	0.004	0.947	1.029	0.445	2.382
Employment opportunities for skilled professionals	-0.002	0.429	0.000	0.997	0.998	0.431	2.313
Threat to the Common Workforce	-0.058	0.373	0.024	0.877	0.944	0.454	1.960
Risk and safety issues for human resources	0.160	0.504	0.101	0.751	1.174	0.437	3.153
Encroachment on air space and visual incursions	0.044	0.408	0.012	0.914	1.045	0.469	2.327
Constant	3.113	1.592	3.824	0.051	22.489		

The logistic regression results for social, legal, and environmental factors (Table 8) demonstrate that mitigating risk and safety issues for on-site personnel positively influences drone adoption. With a regression coefficient (B) of 0.160 and an odds ratio (Exp(B)) of 1.174, the findings suggest a 17.4% increase in adoption probability, indicating that the perceived safety improvements offered by drones are appreciated, albeit not a dominant driver. Concerns related to airspace encroachment and visual disturbances register a B value of 0.044 and an odds ratio of 1.045, resulting in a 4.5% increase in likelihood, suggesting these are relatively minor concerns. Likewise, the impact of government policies and regulations is minimal, with a B coefficient of 0.041 and an Exp(B) of 1.041, showing only a 4.1% increase, which implies that current regulatory measures neither significantly promote nor inhibit adoption. Additionally, the requirement to obtain a drone piloting license, reflected by a B value of 0.029 and an odds ratio of 1.029, contributes to a 2.9% increase in adoption likelihood, indicating that licensing protocols are not perceived as a major obstacle.

On the other hand, organizational acceptance of drone technology presents a considerable barrier. This is evidenced by a B value of -0.862 and an odds ratio of 0.422, translating to a 57.8% decrease in the likelihood of adoption, suggesting notable resistance at the organizational level, possibly due to doubts about the utility or cost-effectiveness of drones. Perceived threats to traditional workforce roles also slightly discourage adoption, with a B coefficient of -0.058 and an odds ratio of 0.944, equating to a 5.6% reduction in adoption probability, indicating mild concern over job displacement. Meanwhile, potential job opportunities for skilled drone professionals show virtually no effect on adoption, with a B value of

-0.002 and an odds ratio of 0.998, leading to a 0.2% decline, implying that employment-related impacts either positive or negative are not influential in decision-making regarding drone usage.

Construction-related Factors

According to the logistic regression analysis of construction-related factors (Table 9), effective stocktaking, monitoring, and control functions offered by drones significantly support their adoption. This is demonstrated by a regression coefficient (B) of 0.492 and an odds ratio (Exp(B)) of 1.636, indicating a 63.6% increase in the likelihood of adoption, highlighting the operational advantages drones bring to construction site management. Similarly, the provision of accurate and high-quality data by drones stands out as a major contributing factor, with a B value of 0.660 and an Exp(B) of 1.934, reflecting a 93.4% rise in adoption probability, which confirms the value placed on precise information as a key adoption driver. In addition, deploying materials, equipment, and workforce through drone assistance has a B value of 0.088 and an odds ratio of 1.092, suggesting a modest positive effect of 8.8%, while collecting data from hard-to-reach locations shows a strong influence with a B value of 0.815 and an Exp(B) of 2.143, enhancing the likelihood of adoption by 114.3%. Although this impact is substantial, it is still considered slightly less important than the value of data accuracy.

On the contrary, some aspects show a negative correlation with adoption. The function of providing essential data and information, despite its utility, is associated with a B coefficient of -0.080 and an odds ratio of 0.923, resulting in a 7.7% decline in likelihood, which may indicate that this capability alone does not serve as a compelling adoption factor. Additionally, the real-time surveillance feature for on-site crime prevention records a B value of -0.345 and an Exp(B) of 0.708, corresponding to a 29.2% reduction in adoption probability, implying that security monitoring is not widely regarded as a priority function when considering drone implementation in construction.

Table 9 Construction Demand related Factors - Logistic Regression Analysis

	B	S.E.	Wald	Sig.	Exp(B)	95% C.I.	
						Lower	Upper
Drones provide critical data and information	-0.080	0.454	0.031	0.860	0.923	0.379	2.250
Effective stocktaking, control, and monitoring	0.492	0.445	1.223	0.269	1.636	0.684	3.911
Real-time surveillance capabilities for crime prevention	-0.345	0.650	0.282	0.596	0.708	0.198	2.533
Deployment of material and equipment	0.088	0.471	0.035	0.852	1.092	0.434	2.747
Collection of data from inaccessible space	0.815	0.577	1.992	0.158	2.143	0.143	1.373
Data Accuracy & High-quality information	0.660	0.516	1.634	0.201	1.934	0.703	5.320
Constant	1.510	2.381	0.402	0.526	4.526		

5. Conclusions

Drones are one of the widely adopted digital technology in the world. The applications are globally explored in various industries like military, agriculture, commercial and the construction industry. In Indian context drones are Majorly explored in Military and agriculture industry. After 2018, various authors have research about the applications and challenges helps to adopt drones in Indian construction industry. However, there is a gap between theoretical research and practical implementations. Figure 1 depicts how this research helps to identify the key success factors impacting drone technology adoption.

To maximize the cost-saving potential of drones, their use should be standardized across all project phases, leading to significant benefits such as 50-60% labour savings, 60-80% time savings, a 30% reduction in material wastage, and lower logistics expenses. Furthermore, increasing drone applications in various construction activities will help reduce costs associated with data downloading, analysis, and interpretation, which previously constituted nearly half of the total project investment. In addition to cost efficiency, improving the availability of drone equipment is essential for widespread adoption, which can be achieved by leveraging the Drone Production-Linked Incentive (PLI) Scheme to boost domestic manufacture and reduce dependence on imports. Alongside equipment availability, addressing skill gaps is equally crucial, necessitating DGCA-endorsed pilot training and certification programs, along with periodic renewals, to ensure standardization and compliance. Moreover, the NaMo Drone Skill Development Scheme can subsidise training costs and provide employment incentives, thereby expanding the skilled workforce required for effective drone operations in construction. Beyond economic and operational aspects, social, legal, and environmental considerations play a significant role

in drone adoption. Streamlined government regulations under Drone Rules 2021 have simplified certification processes and established clear legal frameworks for safe usage. However, addressing safety challenges for human resources remains essential, requiring the implementation of comprehensive DGCA Safety Protocols, including training, risk assessments, and standard operating procedures. Additionally, privacy concerns and airspace management issues can be mitigated through designated No-Fly Zones and the use of the Digital Sky Platform for airspace mapping to establish safe drone corridors. Furthermore, drones enhance construction efficiency by supporting tasks like automated stocktaking, real-time progress monitoring, and inventory management through AI-driven technologies outlined in the National Geospatial Policy 2022. In scenarios where data collection in inaccessible or hazardous areas is necessary, drones equipped with advanced sensors ensure comprehensive and safe data gathering. Moreover, advanced LiDAR and photogrammetry technologies, supported by the Digital India Land Records Modernization Program (DILRMP), play an important role in generating high-quality and precise data for infrastructure projects. These tools not only improve data accuracy but also address the complexities of modern construction demands, ultimately revolutionizing the industry through strategic integration of cost-effective policies, robust training programs, streamlined regulations, and advanced data technologies.

To accelerate the adoption and application of drone technology in the Indian construction industry, future research should focus on several key areas. First, expanding applications across the construction lifecycle is essential, with studies needed to explore drone usage beyond traditional tasks like surveying and monitoring. Potential areas of interest include facility management, demolition, refurbishment, and post-construction phases to maximize their utility across the entire building lifecycle. Additionally, integrating drones with emerging technologies such as Artificial intelligence (AI), Digital Twins, and the Internet of Things (IoT) requires further investigation to enhance automation, data processing, and real-time decision-making. Furthermore, research should examine the applications of drones in promoting sustainable practices by monitoring carbon emissions, optimizing resource efficiency, and implementing eco-friendly construction methods to address global environmental challenges. Finally, conducting comprehensive cost-benefit analyses is crucial to compare traditional and drone-based methods across various project sizes and complexities, providing empirical evidence to justify broader adoption. By addressing these research areas, the construction industry can fully achieve the potential of drone technology, resulting in increased efficiency, sustainability, and creativity.

6. Future scope

To advance the adoption and usage of drone technology in the Indian construction industry, future research should explore several key areas. First, there is a need to expand drone applications across the entire construction lifecycle, moving beyond conventional tasks such as surveying and monitoring. This includes investigating their potential in facility management, demolition, refurbishment, and post-construction phases. Additionally, studies should examine how drones can be effectively integrated with emerging technologies like Digital Twins, artificial intelligence (AI), and the Internet of Things (IoT) to enhance operational efficiency and data-driven decision-making. Research should also focus on the

sustainability and environmental impact of drones, particularly their role in monitoring carbon emissions, improving resource efficiency, and supporting eco-friendly construction practices to address pressing environmental concerns. Lastly, comprehensive cost-benefit analyses are essential to compare traditional methods with drone-based approaches across various project types and scales, providing empirical justification for broader industry adoption.

Declarations

Acknowledgements: Not applicable.

Conflicts of Interest: The authors declare no conflicts of interest.

Funding: The authors declare that no funding, grants, or other support were received during the preparation of this manuscript.

Authors' Contributions: S.P. and L.J. contributed to the study conception and design. S.P. wrote the sections of the manuscript. L.J. reviewed and improved the written sections. All authors contributed to manuscript revision, read, and approved the submitted version. (S.P. – Sivaraman.P & L.J. – Luke Judson)

Competing Interests: The authors declare no competing interests.

Ethics Approval: The need for ethical approval for this questionnaire study was waived by the Ethics Committee of the School of Planning and Architecture, New Delhi, India, due to the nature of the study, which involved minimal risk to participants. All participants provided informed consent prior to their inclusion in the study.

Consent to Participate: All participants involved in the survey and interviews were informed about the purpose of the study and participated voluntarily. Informed consent was obtained from all individual participants included in the study. Participants were assured of confidentiality, and their responses were anonymized prior to analysis. This study complies with ethical standards and guidelines for research involving human subjects.

Consent to Publish: Not applicable.

Data Availability: The data will be available upon reasonable request from the corresponding author.

Clinical Trial Number: Not applicable.

References

1. Aiyetan, A. O. & Das, D. K., 2023. Use of Drones for construction in developing countries: barriers and strategic interventions. *International Journal of Construction Management.*

2. Albeaino, G. & Gheisari, M., 2021. Trends, benefits, and barriers of unmanned aerial systems in the construction industry: a survey in the United States. *Journal of Information Technology in Construction.*
3. Alziadi, T. & Qasem, A., 2022. Role of advanced technologies in enhancing construction management of mountainous road projects in the UAE. *Heriot-Watt University*.
4. Ashour, A., Barakat, A. & Elbeltagi, I., 2022. Orthomosaic applications in drone-based construction supervision. *Journal of Construction Engineering, 28(2), pp. 141–156.*
5. Charlesraj, V. & Rakshith, N., 2020. Stakeholder Perspectives on the Adoption of Construction Projects. s.l., s.n.
6. Ciampa, E., De Vito, L. & Pecce, M. R., 2016. Practical issues on the use of drones for construction inspections. *Journal of Construction Technology, 15(2), pp. 201–209.*
7. Cooper, R. G. & Kleinschmidt, E. J., 1995. Benchmarking the firm's critical success factors in new product development. *Journal of Product Innovation Management, 12(5), pp. 374–391*.
8. DGCA, 2018. Requirements for Operation of Civil Remotely Piloted Aircraft Systems (RPAS). *Directorate General of Civil Aviation, India.*
9. Dron, A. & Dron, B., 2018. Drone Applications in Construction: Budget and Feasibility Aspects. *Journal of Advanced Construction Studies, 5(2), pp. 112–118.*
10. Dua, S. & Kumar, K., 2022. Fire Risk Analysis for Construction Sites. *International Journal of Architectural Design and Management.*
11. Edulakanti, S. R. & Ganguly, S., 2023. The emerging drone technology and the advancement of the Indian drone business industry. *Journal of High Technology Management Research.*
12. Elghaish, F., Matarneh, S. & Talebi, S., 2020. Toward digitalization in the construction industry with immersive and drones technologies: a critical literature review. *Smart and Sustainable Built Environment.*
13. EY, 2022. Making India the drone hub of the world. s.l.: s.n.
14. Falorca, J. F., Miraldes, J. P. N. D. & Goncalves Lanzinha, J. C., 2021. New trends in visual inspection of buildings and structures: Study for the use of drones. *Open Engineering*.
15. Fan, J. & Saadeghvaziri, M. A., 2019. Applications of Drones in Infrastructures and Opportunities. s.l.: s.n.
16. Giordan, D., Manconi, A. & Allasia, P., 2020. Drone imagery for volumetric analysis in construction. *Engineering Geology, 274, p. 105737.*
17. Gulshan Taj, M. N. A. et al., 2021. Monitoring and Management of Construction Sites Using Drone. Proceedings of International Conference on Innovative Technologies for Clean and Sustainable Development.
18. Howard, L., 2023. Labor Safety Insights with Drones in Construction. *Safety Tech Journal, 10(1), pp. 55–62*

19. Irizarry & Costa et al., 2022. Employment opportunities in drone-integrated construction processes. *Journal of Civil Infrastructure, 34(1), pp. 89–95.*
20. Irizarry et al., 2016. Automation and material tracking using drone-based RFID technology. *Construction Technology International, 19(4), pp. 56–65.*
21. Jacobsen, E. L. & Teizer, J., 2022. Deep Learning in Construction: Review of Applications and Potential Avenues. *Journal of Computation in Civil Engineering.*
22. Judson, L. & Paul, V. K., 2019. Uncertainty Factors Affecting Construction Project Cost. SPACE, *The SPA Journal of Planning and Architecture, 23(3-4), pp. 1–17.*
23. Kaamin, M., Fahmizam, M. A. & Jefri, A. S., 2023. Progress Monitoring at Construction Sites Using UAV Technology. s.l.: s.n.
24. Kumarapu, K., Shashi, M. & Keesara, V. R., 2020. UAV in Construction Site Monitoring and Concrete Strength Estimation. *Journal of the Indian Society of Remote Sensing.*
25. Li, Y. & Liu, C., 2018. Applications of multirotor drone technologies in construction management. *International Journal of Construction Management.*
26. Liang, H., Lee, S. C. & Kim, J., 2023. Towards UAVs in Construction: Advancements, and Future Directions for Monitoring and Inspection. *Drones.*
27. Lin, P., Zhang, X. & Liu, M., 2018. Thermal Drone Imaging for Post-Construction Assessment. *Journal of Civil Engineering Technology, 17(4), pp. 401–410.*
28. Lu, M., Wu, Z. & Zhao, Y., 2019. Understanding predictors of drone adoption using logistic regression. *Technology Adoption Quarterly, 7(1), pp. 35–49.*
29. Luthra, S., Mangla, S. & Qureshi, M., 2021. Critical success factors for sustainability initiatives in supply chains. *Journal of Cleaner Production, 310, pp. 127446.*
30. Mahajan, G., 2021. Applications of Drone Technology in Construction Industry: A Study 2012–2021. *International Journal of Engineering and Advanced Technology.*
31. Marzappour, L., 2019. UAV-assisted site inspection in high-rise building projects. Smart Infrastructure Magazine, 6(3), pp. 119–127.
32. Martinez, J. G. & Gheisari, M., 2020. UAV Integration in Current Construction Safety Planning and Monitoring Processes: Case Study of a High-Rise Building Construction Project in Chile. J. Manage. Eng.
33. Mashari, M., Al-Mudimigh, A. & Zairi, M., 2003. Enterprise resource planning: A taxonomy of critical factors. European Journal of Operational Research, 146(2), pp. 352–364.
34. McNabb et al., 2017. Drone compliance and regulation for construction safety. Journal of Safety Engineering, 18(2), pp. 142–150.
35. Moza, A. & Paul, V., 2024. Critical Delay Factors Affecting Construction Project Performance – A Contemporary Perspective. European Project Management Journal.
36. Nachiappan, B. et al., 2024. Exploring the Application of Drone Technology in the Construction Sector. s.l.: s.n.

37. Nousi, A., Lefas, I. & Petrou, M., 2019. Orthomosaic assessment in construction management. Journal of Remote Construction Sensing, 11(1), pp. 65–76.
38. Patel, R., 2021. Workforce surveillance with UAVs: A safety enhancement study. International Safety Engineering Journal, 20(4), pp. 320–329.
39. Patil, S. S., Admuthe, V. S. K. & Patil, R. K., 2022. Construction site inspection by using Drone or UAV. International Journal of Engineering Applied Sciences and Technology.
40. Patterson, J., 2018. Site selection analytics using UAVs. Journal of Construction Technology, 14(3), pp. 211–223.
41. PWC, 2023. Building trust in Commercial drones. s.l.: s.n.
42. Raj, A. & Sah, B., 2019. Analyzing Critical Success Factors for Implementation of Drones in the Logistics Sector Using Grey-DEMATEL based approach. s.l.: s.n.
43. Rajagopalan, R. P. & Krishna, R., 2018. Drones: Guidelines, Regulations, and Policy Gaps in India. s.l.: s.n.
44. Rao, A. S., Radanovic, M. & Liu, Y., 2022. Real-time monitoring of construction sites: Sensors, methods, and applications. Automation in Construction.
45. Ruiz, M., Delgado, A. & Perez, C., 2021. Photogrammetry for on-site fabrication analysis using UAVs. Construction Imaging Review, 16(1), pp. 87–95.
46. Saini, A., Khursheed, S. & Kumar, K., 2021. Analysis of Indirect Cost of Construction Projects in India. International Journal of Sustainable Building Technology.
47. Sankalpa, H. L. C., 2020. Investigating the usage potential of Drones/ Unmanned Aerial Vehicle (UAV) in Construction Industry. ResearchGate.
48. Shakhatreh, H. & Sawalmeh, A. H., 2019. Unmanned Aerial Vehicles (UAVs): A Survey on Civil Applications and Key Research Challenges. IEEE Access.
49. Shankar, R., Yadav, S. & Kannan, D., 2023. Blockchain and IoT for Traceability in Food Logistics Systems. Computers & Industrial Engineering, 167, p. 108011.
50. Siebert, S. & Teizer, J., 2022. Cost-benefit analysis of drone deployment in construction sites. Automation in Construction, 130, p. 103848.
51. Simon, D., Paul, V., Dua, S. & Kumar, K., 2022. Fire Risk Analysis for Construction Sites. International Journal of Architectural Design and Management.
52. Sobola, T. M., 2021. UAVs in Land Use Planning and Zoning Analysis. Urban Planning Today, 9(4), pp. 90–98.
53. Sridharan, S. R., Malsane, S. & Bhutada, G. S., 2023. Adoption of drone technology in construction – a study on interaction between various challenges. World Journal of Engineering.
54. Stocker, C., Nex, F. & Koeva, M., 2020. High-Quality UAV-Based Orthophotos for Cadastral Mapping: Guidance for Optimal Flight Configurations. Remote Sensing.
55. Tasevski, Z., Aleksic, D. & Markovic, M., 2018. UAV project closeout analysis in infrastructure projects. Journal of Construction Management, 19(2), pp. 155–162.

56. The Drone Rules, 2021. The Drone Rules - Realising our collective vision of an Aatmanirbhar Bharat. Ministry of Civil Aviation.
57. TreckR, G., Mendes, R. & Liu, H., 2018. UAV progress tracking in large-scale construction. Automation in Construction, 92, pp. 254–262.
58. Tutas, M., Kovacs, I. & Bela, T., 2021. Improving Construction Surveillance with UAVs. Proceedings of the Construction Monitoring Conference.
59. Tutz, G., 2022. Regression for Categorical Data. Cambridge University Press.
60. Vanathi & Radhika, 2022. Rise of Drones in Indian construction industry. International Research Journal of Engineering and Technology.
61. Vergouw, B., Nagel, H. & Duin, R., 2022. The future of drones in facility inspections. Drone Technology Review, 6(2), pp. 33–45.
62. Vishwakarma, K., Paul, V. & Solanki, S., 2022. Analysis of Key Factors Affecting Labour Productivity in Building Rehabilitation Projects. International Journal of Sustainable Building Technology.
63. Williams, R. & Quiroz, M., 2020. Logistic regression models for predicting technology adoption. Statistical Modeling for Social Sciences, 18(1), pp. 23–40.
64. Wingtra, 2021. Drone Mapping Applications in Construction: Global Insights. Wingtra White Paper Series.
65. Yi, L., Zhang, T. & Chen, Q., 2020. Site activity monitoring using drones and computer vision. Journal of Construction Automation, 22(3), pp. 198–210.
66. Zhang, C., Wang, F. & Zou, Y., 2023. Automated UAV image-to-BIM registration for building facade inspection using improved generalised Hough transform. Automation in Construction.
67. Zhang, Y., et al., 2019. Economic and operational assessment of drone integration in modern construction sites. Journal of Engineering Technology, 25(3), pp. 178–192.
68. Zhou, W. & Gheisari, M., 2021. Drone Training Requirements and Capacity Building in Construction. Journal of Safety and Health in Construction, 13(2), pp. 140–149.

Figures

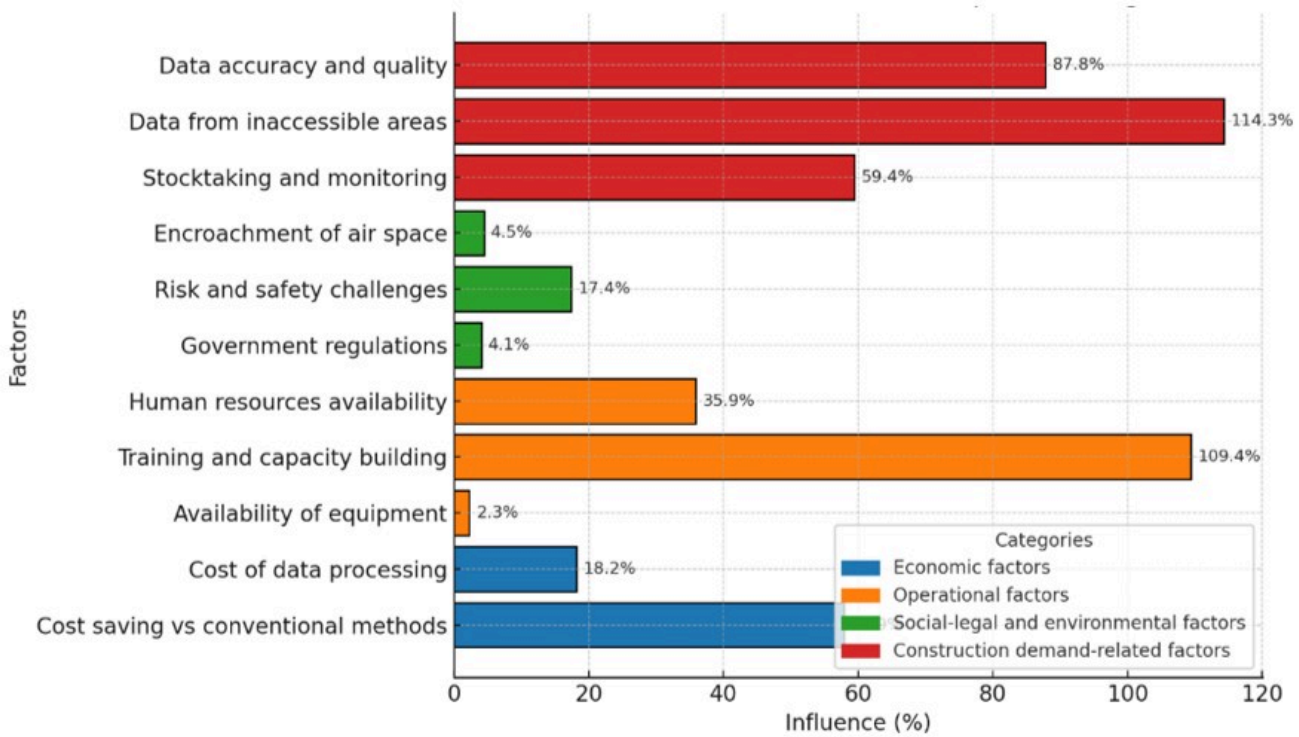

Figure 1

Influencing factors on drone adoption (Source: Author)